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Design and Development of Automated Mobile Robotic Arm Vehicle for Cold Storage Operations

Presented By

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Project Guide

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Introduction

•Cold storage warehouses face challenges like -20°C temperatures.
•Human labor becomes inefficient.
•Need for mobile, autonomous robotic systems.
•Solution: A robotic arm vehicle for cold storage tasks.

Gaps in Knowledge:

Although many studies focus on robots working in challenging environments, there is limited research on mobile robotic systems specifically designed to operate in cold storage. Most robotic systems are not built to handle the combination of mobility, precision, and durability required in such low-temperature environments.

Literature Survey Highlights

•Fixed AS/RS systems lack adaptability.

•Mobile robots need insulation/sensors.

•AI & LiDAR effective but require modifications for cold.

•Real-time environmental adaptation critical.

Problem Identification

- Challenges:
  - Mechanical failures at low temps
  - Sensor fogging
  - Battery inefficiency
- Limitations in existing robotics
- Need for mobile manipulation in cold storage

System Development Phases

- Research & Requirements
- Design & Architecture
- Prototyping & Integration
- Testing & Optimization
- Deployment

Project Setup

Components:

- Servo Motors

- Servo Controller

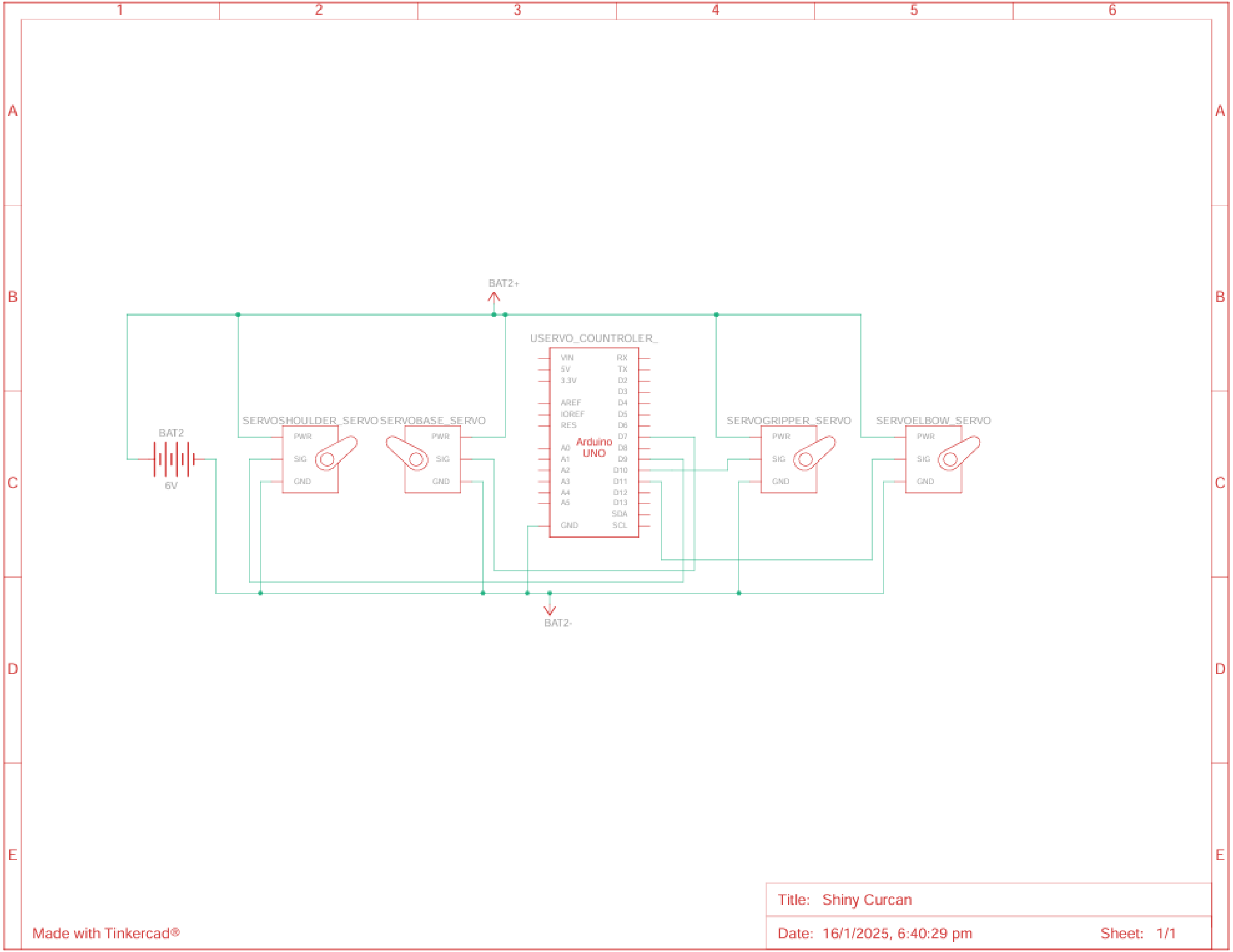
- Battery

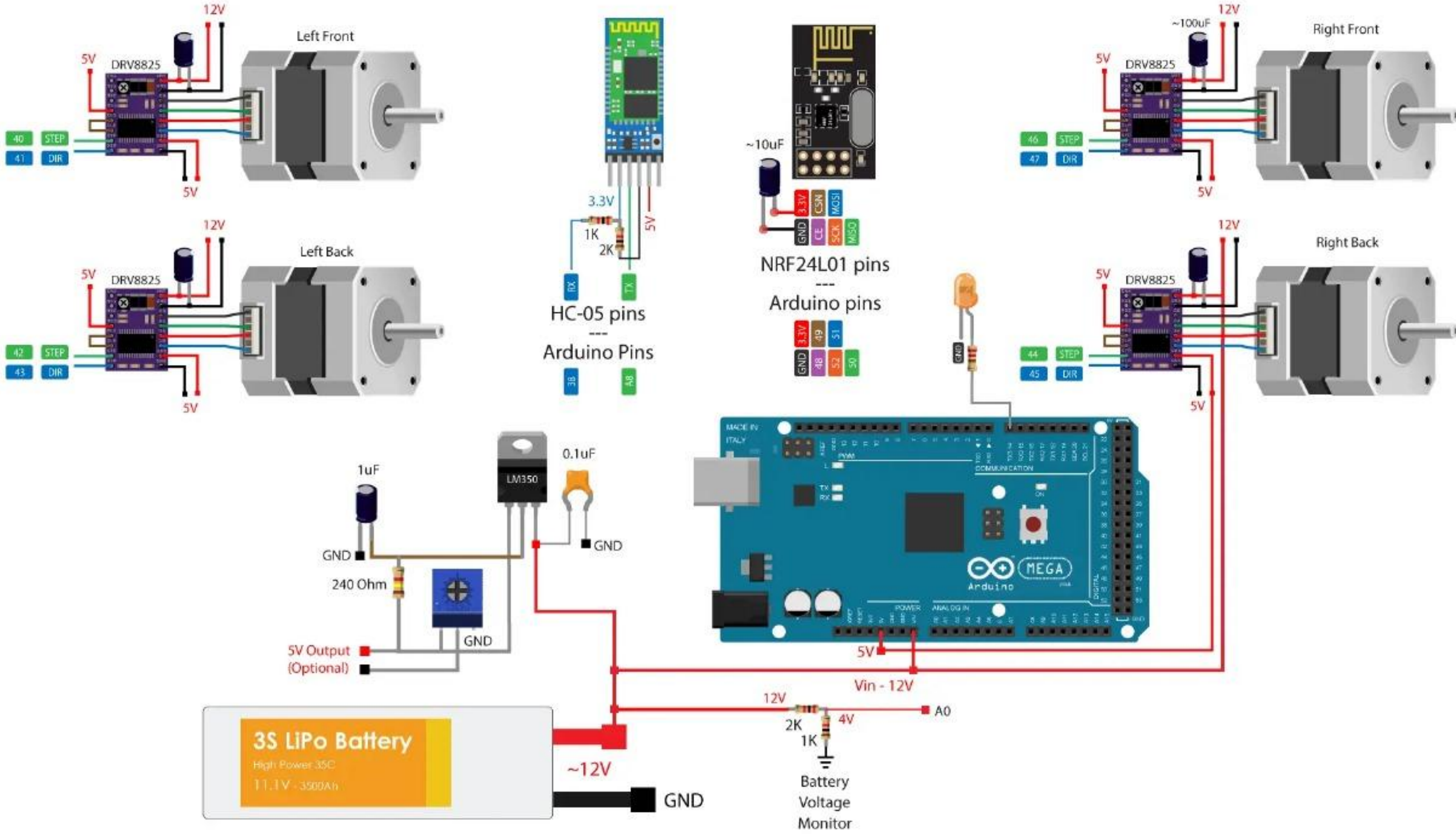
- EEZYbotARM MK2 (3D printing in progress)

-Stepper Motor

-DRV8825 Stepper Motor Driver Module (High Current)

-ControllersTech HC-05 master-slave 6pin Bluetooth Transceiver Module


BAT2+
USERVO_COUNTROLER_
VIN
5V
3.3V
AREF
IOREF
RES
A0
A1
A2
A3
A4
A5
GND
RX
TX
D2
D3
D4
D5
D6
D7
D8
D9
D10
D11
D12
D13
SDA
SCL
Arduino UNO
SERVOSHOULDER_SERVO
SERVOBASE_SERVO
SERVOGRIPPER_SERVO
SERVOELBOW_SERVO
PWR
SIG
GND
BAT2
6V
BAT2-
Title: Shiny Curcan
Date: 16/1/2025, 6:40:29 pm
Sheet: 1/1
Made with Tinkercad®



Left Front
Left Back
Right Front
Right Back
DRV8825
12V
5V
~100uF
STEP
DIR
40
41
42
43
44
45
46
47
3.3V
1K
2K
RX
TX
HC-05 pins

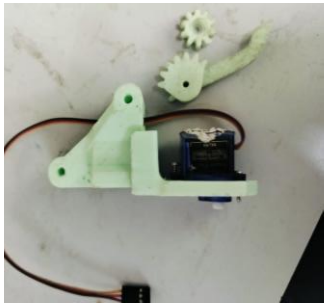
Arduino Pins
38
A8
~10uF
3.3V
CSN
MOSI
GND
CE
SCK
MISO
NRF24L01 pins

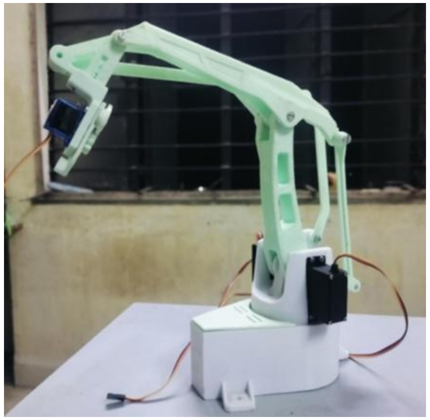
Arduino pins
49
51
48
52
50
GND
1uF
LM350
0.1uF
GND
240 Ohm
5V Output
(Optional)
5V
Vin - 12V
12V
2K
4V
1K
A0
Battery
Voltage
Monitor
3S LiPo Battery
High Power 35C
11.1V - 3500Ah
~12V
GND

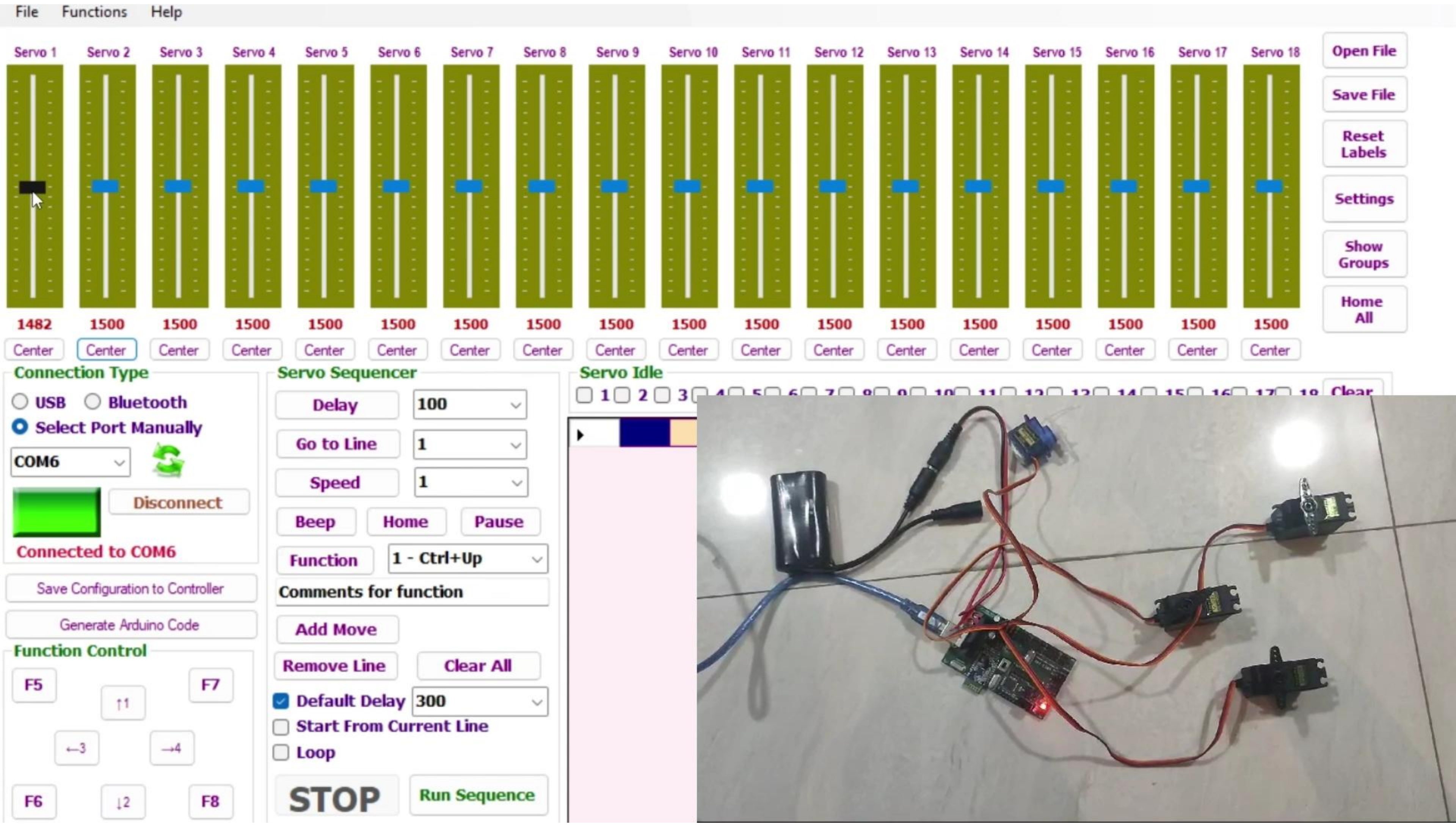
System Design

- 3D printed parts (PLA/ABS)
- Servo motors and gear system
- Parallel linkage for smooth motion
- Omnidirectional base (Mecanum wheels)






File
Functions
Help
Servo 1
Servo 2
Servo 3
Servo 4
Servo 5
Servo 6
Servo 7
Servo 8
Servo 9
Servo 10
Servo 11
Servo 12
Servo 13
Servo 14
Servo 15
Servo 16
Servo 17
Servo 18
Open File
Save File
Reset Labels
Settings
Show Groups
Home All
1482
1500
Center
Connection Type
USB
Bluetooth
Select Port Manually
COM6
Disconnect
Connected to COM6
Save Configuration to Controller
Generate Arduino Code
Function Control
F5
F7
F6
F8
Servo Sequencer
Delay
100
Go to Line
1
Speed
1
Beep
Home
Pause
Function
1 - Ctrl+Up
Comments for function
Add Move
Remove Line
Clear All
Default Delay
300
Start From Current Line
Loop
STOP
Run Sequence
Servo Idle
Clear

Features

•AI-driven object recognition

•LiDAR-based mapping

•Cold-resistant components

•Autonomous and manual control options

•Adaptive gripper and navigation

Pick and Place Operation

- Object detection using AI & LiDAR
- Navigation and positioning
- Gripping and lifting
- Transport to destination
- Release and reset

Analysis & Calculations

- Payload: 0.26 kg with safety factor
- Torque: 98.1 N.cm
- Motor Power: 2.05 W
- Ensures energy efficiency & reliability

Conclusion

- Effective automation for cold storage
- Combines mobility, AI, and precision
- Future scope: IoT integration, advanced AI, real-time monitoring

THANK YOU